SatuTe: branch-specific detection of phylogenetic signal

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Abstract

Sequence saturation can erase the historical signal across a branch even when that branch has high conventional support. SatuTe addresses this problem directly by testing whether the subtrees separated by a branch remain statistically coherent. Using the original SatuTe test as the primary model, we first reproduce its Jukes–Cantor benchmark and show that bootstrap support can remain complete after branch-specific signal has fallen to the null rejection rate. We then evaluate two model-based extensions: an eigenvalue-weighted statistic that incorporates all persistent non-stationary directions of a reversible substitution model, and a corrected null for invariant-site mixtures. The weighted statistic reduces exactly to original SatuTe under Jukes–Cantor and improves detection mainly for short alignments near the long-branch decision boundary. Its largest matched GTR+F+G4 gain was 57.6 percentage points under taxon Bonferroni correction; across 1,728 matched protein cells, 831 differences were positive and none were negative. Independent truth-labeled experiments supported false-discovery control. Applications to Tree-of-Life and hepatitis C virus data localized broad losses of signal in 16S rRNA and codon-position-specific losses in viral coding regions. These results establish SatuTe as a complement to branch support and show how model-aware extensions can refine, without replacing, its branch-specific diagnosis of phylogenetic signal.

Model-based phylogenetics combines tree search with branch-support measures such as the bootstrap and approximate likelihood-ratio tests [1, 2, 3, 4]. These measures quantify the reproducibility of, or local preference for, a split. They do not test whether repeated substitutions have erased the historical dependence connecting the groups on either side of that split. A branch can therefore receive high support even when the alignment no longer contains recoverable signal across it.

Saturation is the progressive loss of historical information as multiple substitutions accumulate at the same sites [5, 6]. Alignment-wide indices and locus filters can identify heavily saturated data [7, 8], but deep phylogenies often contain mixtures of informative and uninformative genes, regions, and rate categories. A branch-specific diagnostic is needed to localize this loss.

SatuTe generalizes saturation from pairs of sequences to the two subtrees separated by a branch [9]. Its null hypothesis is that the subtrees evolve independently. Rejection indicates detectable shared history; failure to reject classifies the branch as saturated at the selected threshold. SatuTe therefore complements branch support: branch support asks whether a split is reproducible, whereas SatuTe asks whether the alignment still connects its two sides.

For a site pattern ∂, partial likelihood vectors $L(\partial_A)$ and $L(\partial_B)$ give the probabilities $P(\partial_A) = \langle L(\partial_A), \pi \rangle$ and $P(\partial_B) = \langle L(\partial_B), \pi \rangle$ for the subtrees separated by the focal branch. Under the spectral decomposition of a reversible substitution model [10, 11], with eigenvalues $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \cdots$ and left eigenvectors h_i, the coherence coefficient is

$$C_{\partial i} = \left\langle \frac{L(\partial_A)}{P(\partial_A)}, h_i \right\rangle \left\langle \frac{L(\partial_B)}{P(\partial_B)}, h_i \right\rangle . \tag{1}$$

Original SatuTe retains the slowest non-stationary component. This published formulation remains the reference test throughout our analyses.

We first reproduce the original Jukes–Cantor (JC) design and use it to distinguish branch support from recoverable historical signal. We then ask whether faster spectral components should contribute while they persist on a finite branch and derive weights from their decay

relative to the slowest component. We also correct the infinite-branch null for explicit invariant sites, which retain ancestral dependence even as the focal branch tends to infinity. Both changes preserve the branch-specific question posed by SatuTe rather than define a different primary model.

We evaluate these extensions in paired nucleotide and protein simulations under three decision rules and in independent truth-labeled calibration experiments. Finally, we apply original and extended SatuTe analyses to ribosomal-protein two-domain and 16S rRNA three-domain Tree-of-Life examples [12, 13, 14], and to codon-position partitions from a hepatitis C virus reference panel. These applications are designed to identify branches and alignment regions that retain detectable signal under a fitted model, not to select a Tree-of-Life topology or infer viral population structure.

Results

Original SatuTe defines the branch-specific test

SatuTe tests whether the subtrees separated by a focal branch retain detectable statistical dependence. For the slowest non-stationary component, original SatuTe averages the coherence coefficient in Eq. (1) across n independently evolving sites and standardizes that mean with its estimated null variance. The resulting one-sided statistic rejects subtree independence when it exceeds the selected normal threshold. When the tree is inferred from the same alignment, SatuTe applies a taxon-pair Bonferroni threshold to account for topology selection.

This branch, null hypothesis, and decision rule define the primary model used here. The extensions below alter how coherence is summarized under models with distinct non-stationary eigenvalues or explicit invariant sites; they do not change the biological question posed by SatuTe.

A weighted coefficient extends the SatuTe statistic

The spectral decomposition underlying SatuTe gives

$$\frac{P(\partial \mid t)}{P(\partial_A)P(\partial_B)} = 1 + \sum_{i=1}^{d} C_{\partial i} \exp(\lambda_i t). \tag{2}$$

Here d is the number of non-stationary eigenvalues: $d = 3$ for DNA and $d = 19$ for an amino-acid model. The coefficient $C_{\partial i}$ measures agreement between the two subtrees in spectral direction h_i, whereas $\exp(\lambda_i t)$ measures how much of that direction persists across the focal branch. Because all non-stationary eigenvalues are negative, components with more negative λ_i decay faster.

For pattern ∂, we define the eigenvalue-weighted coherence as

$$C_{\partial}^{\lambda}(t) = \sum_{i=1}^{d} w_i(t) C_{\partial i}, \qquad w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}. \tag{3}$$

The relative weight $w_i(t)$ sets $w_1(t) = 1$ and downweights components that decay faster than the slowest non-stationary component. Original SatuTe therefore remains the reference scale, while additional components contribute according to the fitted branch length.

For an alignment of n independently evolving sites, the empirical coefficient is

$$\hat{C}_{\lambda}(t) = \frac{1}{n} \sum_{s=1}^{n} C_{\partial_s}^{\lambda}(t). \tag{4}$$

Including the mode weights in the original variance estimator gives

$$\hat{\sigma}_{\lambda}^2(t) = \sum_{i=1}^{d} \sum_{j=1}^{d} w_i(t) w_j(t) \hat{\sigma}_{A,i,j}^2 \hat{\sigma}_{B,i,j}^2. \tag{5}$$

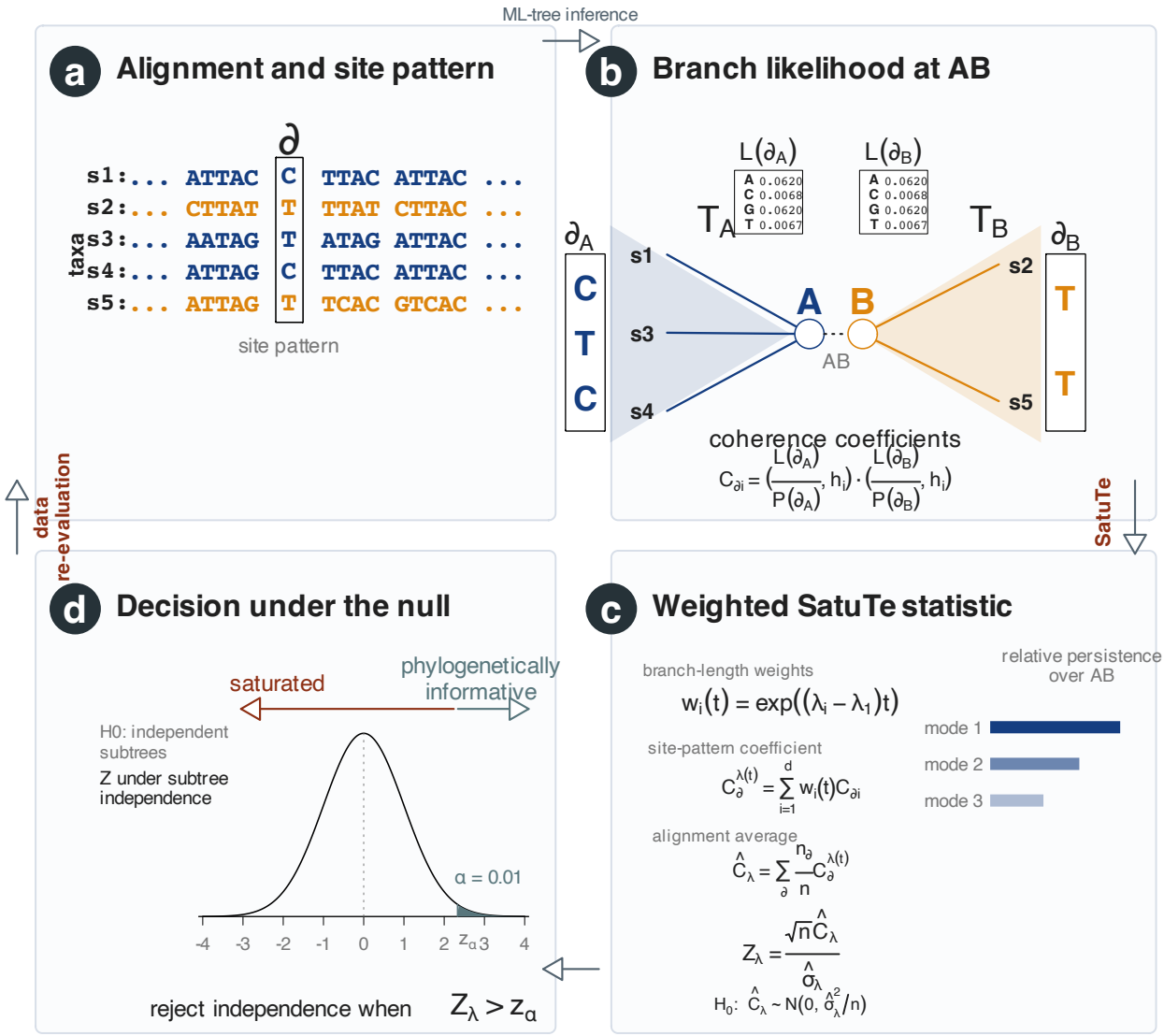


Figure 1: **Branch-specific detection of phylogenetic signal with SatuTe. a,** Columns of a five-taxon alignment define site patterns ∂. **b,** The focal branch partitions each pattern into two subtree patterns used to calculate partial likelihoods and coherence coefficients. **c,** Original SatuTe uses the slowest non-stationary component; the model-aware extension weights additional components by their persistence relative to that reference over branch length t. **d,** Rejection of subtree independence indicates detectable signal, whereas failure to reject indicates saturation at the selected threshold.

The corresponding statistic is

$$Z_\lambda(t) = \frac{\sqrt{n}\hat{C}_\lambda(t)}{\hat{\sigma}_\lambda(t)}. \tag{6}$$

The original decision rule is unchanged: the focal branch is called phylogenetically informative when $Z_\lambda(t) > z_\alpha$, with threshold $z_{\alpha/(m_A m_B)}$ when the tested topology is inferred from the same alignment.

The Jukes–Cantor formulation is unchanged

The JC model has one stationary eigenvalue and a three-dimensional non-stationary eigenspace with $\lambda_1 = \lambda_2 = \lambda_3$. Therefore $w_i(t) = 1$ for all non-stationary modes in Eq. (3), and

$$C_\partial^\lambda(t) = \sum_{k=1}^{3} C_{\partial 1,k} = C_{\partial 1}, \tag{7}$$

which is the multiplicity-three expression used by original SatuTe. The weighted statistic thus makes no change when the substitution model does not distinguish among non-stationary directions.

Invariant categories require a corrected null likelihood

An explicit invariant category is a boundary case of a rate mixture. Let q_c and r_c be the proportion and rate of category c, and let A_{sc} and B_{sc} be the two subtree-pattern probabilities for site pattern s. For a positive-rate category, the infinite-branch null contribution is $q_c A_{sc} B_{sc}$. For the invariant category $c = 0$, however, $r_0 = 0$, and the transition matrix remains the identity even as the focal branch length tends to infinity. Its null contribution is therefore the joint probability

$$J_{s0} = \sum_a \pi_a L_{A,s0}(a) L_{B,s0}(a), \tag{8}$$

rather than the product $A_{s0}B_{s0}$. The corrected mixture null is

$$p_\infty(s) = q_0 J_{s0} + \sum_{c:r_c>0} q_c A_{sc} B_{sc}. \tag{9}$$

For the positive-rate categories, define the model-weighted site score

$$u_s(t) = \frac{\displaystyle\sum_{c:r_c>0} q_c A_{sc} B_{sc} \sum_{i=1}^{d} e^{\lambda_i r_c t} C_{sc,i}}{p_\infty(s)}. \tag{10}$$

The invariant component enters only the denominator and contributes no finite-versus-infinite-branch signal, giving the exact identity

$$\frac{p_t(s)}{p_\infty(s)} = 1 + u_s(t). \tag{11}$$

In computation, all positive-rate persistence weights are divided by the common factor e^δ, where $\delta = \max_{c:r_c>0,i} \lambda_i r_c t$. This prevents underflow without changing the standardized statistic. The refined invariant-null test uses the sample mean and variance of these globally rescaled site scores.

A counterfactual construction that replaces J_{s0} with $A_{s0}B_{s0}$ treats the invariant category as if it became independent across an infinite focal branch. It does not satisfy Eq. (11) under a $+I$ model and need not have null expectation zero.

The original SatuTe benchmark separates support from signal

We retained the original SatuTe simulation controls: five- and 16-taxon trees, alignments of 100–10,000 sites, focal branch lengths of 0.1–10, and 1,000 JC replicates per point. Four analyses separate branch-length estimation from topology selection: the true tree and branch lengths; the true tree with maximum-likelihood (ML) branch lengths; an ML tree and branch lengths; and the same inferred-tree analysis with taxon-pair Bonferroni correction.

With the true tree and branch lengths, the fraction of informative calls declined from nearly one to the 5% null level as the focal branch length increased; longer alignments retained signal across longer branches (Supplementary Fig. S3). Estimating ML branch lengths on the true topology closely followed this reference. Selecting and testing a topology on the same alignment was liberal at long branches, reaching approximately 15% for the external target and 50% for the internal target. Taxon-pair Bonferroni correction largely restored the reference curve, although it became conservative for subtrees containing nearly indistinguishable sequences.

For 100-site, 16-taxon alignments, correct topology recovery fell from 63.2% at focal branch length 0.1 to 4.5% at length 1 and to nearly zero beyond length 3. Bootstrap support for the target split nevertheless reached 100% by length 0.8 and remained there, whereas Bonferroni-corrected SatuTe calls fell to the 3.5–4.6% null range (Supplementary Tables S2 and S3). Thus, high split support does not imply recoverable historical signal across the split.

JC also provides the prespecified formula control. Its three non-stationary directions have equal eigenvalues, so weighted and original SatuTe curves coincide exactly. Differences between the formulations can arise only in models with distinct non-stationary eigenvalues and occurred mainly for short alignments near the long-branch detection boundary.

Extension effects are concentrated on the focal branch

A branch-neighborhood audit showed that weighting did not produce a tree-wide score shift. At branch length 12 and 100 sites, informative-call fractions on the focal branch increased by 14.5–22.0 percentage points under GTR+G4 and by 38.1–39.1 points under LG+G4, whereas changes on adjacent branches were effectively zero (Supplementary Fig. S1).

The corrected invariant null restores false-positive control

We first used exact enumeration to validate the finite-branch $+I$ null. This exploratory $+I+G4$ and $+I+R4$ grid is distinct from the fixed-model publication benchmark below. Across 270 model–branch–alignment-length cells, 10.8 million paired alignments were evaluated. The likelihood identity in Eq. (11) held to 1.55×10^{-11}. The refined score had exact null expectation below 1.53×10^{-16} in absolute value, whereas a counterfactual score that treated invariant sites as independent retained expectations of 0.0354–0.2176.

At the ordinary one-sided $N(0,1)$ threshold, counterfactual rejection averaged 0.900; refined rejection averaged 0.0492. Split-sample empirical thresholds brought held-out rejection near 0.05 for both formulas, but did not make the counterfactual statistic a standard-normal pivot: at $t = 10{,}000$, its threshold rose from 3.15 at 100 sites to 18.47 at 10,000 sites, whereas the refined threshold remained between 1.62 and 1.69.

The refined invariant statistic gains power at the detection boundary

After independent calibration, the refined statistic had the larger exact null-centered separation in 245 of 270 cells and equal separation in the remaining 25. Gains formed a diagonal ridge as longer alignments moved the detection boundary to longer branches (Fig. 2). The maximum $+I+G4$ gain was 0.116 at $t = 60$ and 10,000 sites (bootstrap 95% interval, 0.111–0.122); the maximum exploratory $+I+R4$ gain was 0.163 (0.157–0.170). Supplementary Table S1 reports maxima by alignment length.

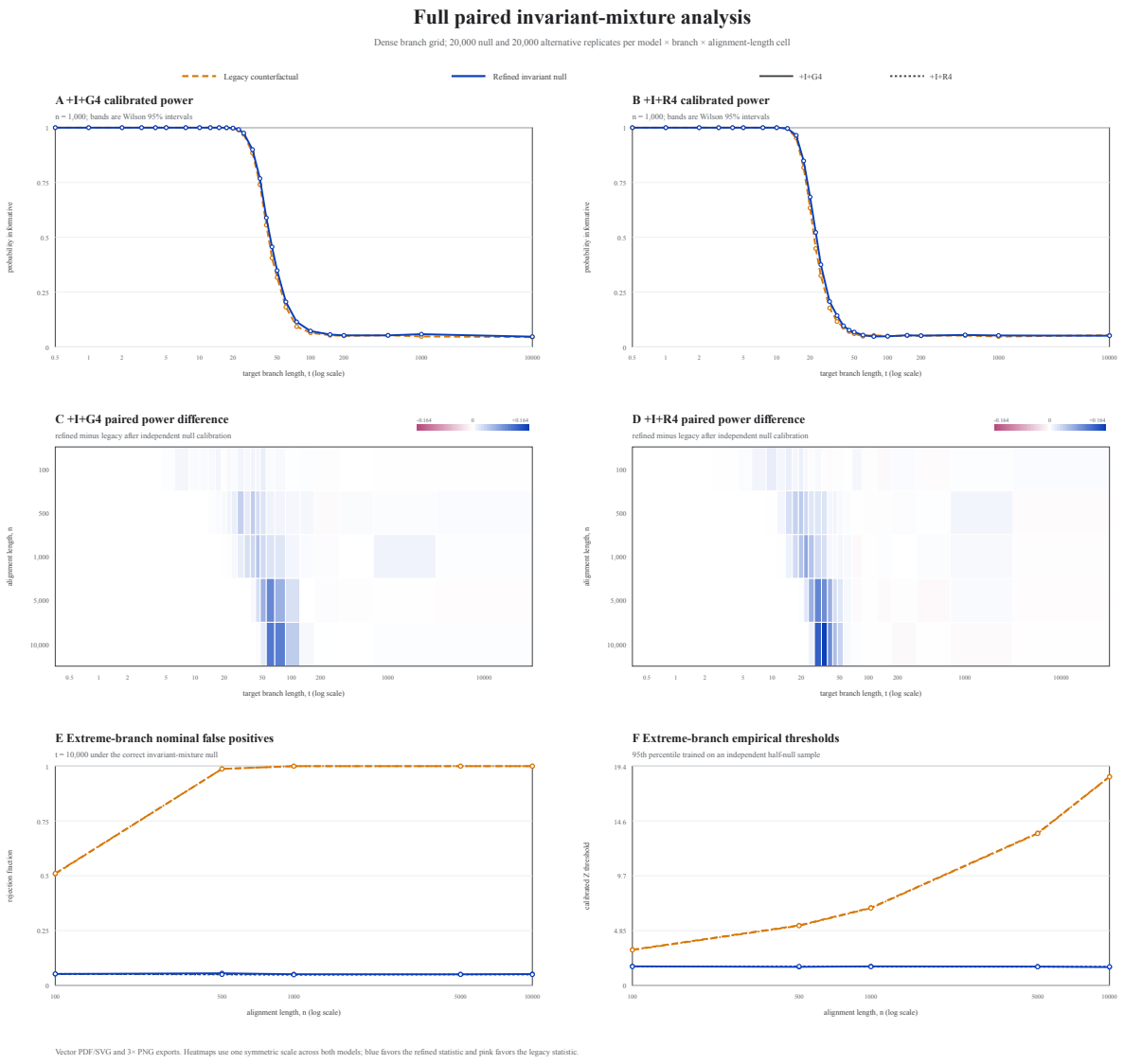


Figure 2: **Correcting the invariant-site null restores calibration.** **a,b,** Empirically calibrated power at 1,000 sites across the branch grid; bands are Wilson 95% intervals. **c,d,** Refined-minus-counterfactual power for five alignment lengths. Blue favors the refined statistic and pink the counterfactual. **e,** Nominal false-positive rates at $t = 10{,}000$. **f,** Empirical 95th-percentile thresholds trained on an independent half-null sample.

Publication-scale nucleotide comparisons

We evaluated fixed GTR+F, GTR+F+G4, and GTR+F+I+G4 models in 1,000 paired replicates per cell, using alignments of 100, 250, or 1,000 sites; focal branch lengths of 4–12; and five-taxon external or 16-taxon internal targets. The merged grid contains 864,000 detail rows and 2,016 summaries, with complete replicate identifiers and no JC-collapse mismatch.

Both GTR+F+G4 formulations lost power as the focal branch length increased, and longer alignments shifted the transition rightward (Fig. 3). Weighting produced its largest shift at 100 sites, where original SatuTe had left the high-power regime but the weighted statistic still detected coherence. Supplementary Figs. S4–S6 report paired differences and the corresponding GTR+F and GTR+F+I+G4 curves.

The largest GTR+F+G4 increase was 0.241 without adjustment (95% bootstrap interval, 0.215–0.269), 0.576 under taxon Bonferroni correction (0.546–0.607), and 0.414 under Benjamini–Yekutieli (BY; 0.384–0.446). All three maxima occurred at 100 sites on the 16-taxon target. The corresponding exact McNemar P values were 5.7×10^{-73}, 8.1×10^{-174}, and 4.7×10^{-125}.

Model misspecification changes absolute operating characteristics

We simulated GTR+F alignments and evaluated each alignment under GTR, JC, K2P, or F81. Across 960,000 detail rows, JC and K2P were more liberal than correctly specified GTR, whereas F81, which retains base composition, was substantially closer (Fig. 4). A small paired difference between formulas under a misspecified model therefore does not imply correct absolute calibration.

Within the misspecified analyses, the largest weighted gains were 0.131 without adjustment, 0.206 under taxon Bonferroni correction, and 0.209 under BY. JC remained an exact formula control because its non-stationary eigenvalues are equal. Supplementary Figs. S11–S13 show JC, K2P, and F81 paired differences and weighted curves.

Matched protein power depends strongly on the model

The matched protein suite used fixed LG+G4, WAG+G4, JTT+G4, and Q.pfam+G4 models. Across 1,728 model–rule–grid cells, weighted-minus-original differences were positive in 831, zero in 897, and negative in none. Maximum taxon-Bonferroni gains were 0.327 for LG, 0.469 for WAG, 0.555 for JTT, and 0.354 for Q.pfam. The complete original, weighted, and paired-difference curves show strong matrix-specific transitions (Supplementary Figs. S7–S10).

Independent truth-labeled calibration supports error control

Independent calibration used 5,000 exact-null replicates per cell, split equally between threshold training and held-out evaluation, plus 2,000 50:50 mixed-null replicates. In the GTR mixed-null design, power ranged from 0.929 to 0.989 and empirical FDR from zero to 0.00386. Formula-paired differences were at most 0.000571 for power and 0.000143 for FDR, confirming that the larger publication-grid effects are localized to the saturation boundary (Supplementary Fig. S14).

Across 96 cells, held-out rejection ranged from 0.0108 to 0.0644; unadjusted and BY GTR cells ranged from 0.0432 to 0.0596. Sixteen-taxon Bonferroni cells were conservative because only complete tied blocks of discrete adjusted P values were attainable.

SatuTe localizes alignment-specific losses in Tree-of-Life data

We revisited two alignments used for deep Tree-of-Life inference [12]: 2,596 amino-acid sites from 16 ribosomal proteins across 3,083 taxa, supporting a two-domain (2D) tree; and 1,947 16S rRNA sites across 1,871 taxa, supporting a three-domain (3D) tree. Branch lengths were

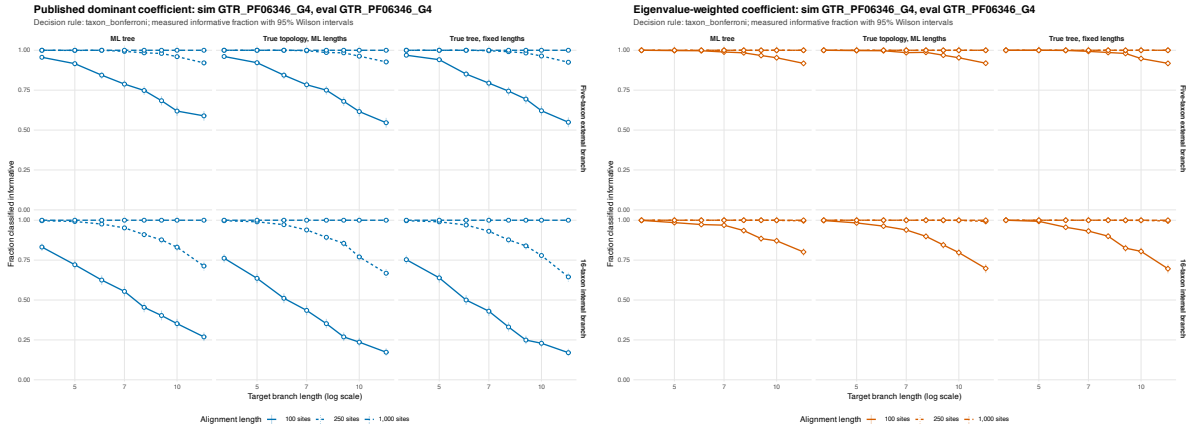


Figure 3: **Matched GTR+F+G4 power.** **a,** Original SatuTe statistic. **b,** Eigenvalue-weighted statistic. Points show informative-call fractions among 1,000 paired replicates under taxon Bonferroni correction; curves are stratified by tree and analysis scenario, and line styles indicate alignment length. Weighting preserves power further into the long-branch regime, particularly at 100 sites.

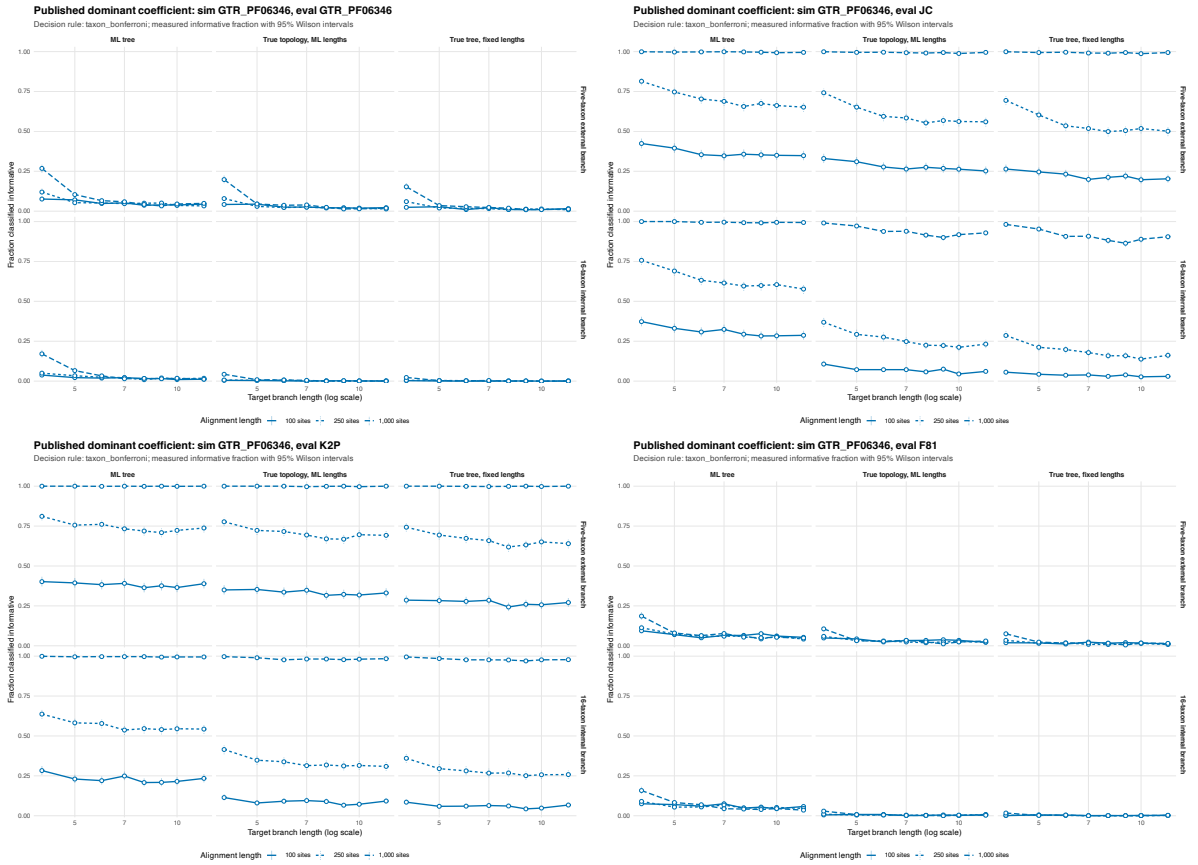


Figure 4: **Model misspecification changes absolute power.** Original SatuTe taxon-Bonferroni curves for GTR data evaluated under GTR (top left), JC (top right), K2P (bottom left), or F81 (bottom right). JC and K2P are more liberal at long branches, whereas F81 approaches the correctly specified analysis.

re-estimated under LG+G4 and GTR+G4, respectively. We tested the internal branch leading to Eukaryota and an external yeast branch (Supplementary Fig. S2).

In the protein 2D tree, original SatuTe classified all 6,163 branches as informative in each of the four rate categories. In the fastest rRNA category (rate 2.73), it classified 107 of 3,739 branches as saturated; 13 were internal, including the Eukaryota branch.

Sliding windows localized this contrast (Fig. 5). Original SatuTe classified 822 of 1,912 rRNA windows (43.0%) as saturated for the Eukaryota branch, compared with 138 of 2,561 protein windows (5.4%). Protein losses were short and often coincided with boundaries in the concatenation. rRNA losses were enriched in hypervariable regions V1–V9 [15]; conserved regions C1 and C4–C9 retained signal, whereas C2 and C3 were nearly saturated. Gappy intervals at positions 240–301 and 789–956 contributed little information.

The weighted sensitivity analysis reduced, but did not erase, this pattern. Native dominant and weighted analyses classified 895 and 699 rRNA windows as saturated for Eukaryota, and 83 and 29 for yeast, respectively. In the protein alignment, original dominant and native weighted analyses classified 138 and 5 windows as saturated for Eukaryota, and 54 and 8 for yeast. Because these cross-implementation comparisons also differ in input handling, the paired simulations provide the controlled estimate of the formula effect.

All 16 ribosomal proteins produced significant scores under both 2D and rearranged 3D placements of Eukaryota; L15, L22, and S10 favored the rearranged placement. Six of nine rRNA regions had larger scores under the 3D placement. The two distributions of score differences did not differ significantly (Kolmogorov–Smirnov $p = 0.072$). These results localize signal for candidate placements rather than select a true topology (Supplementary Tables S4 and S5).

SatuTe exposes codon-position-specific losses in HCV

An independent hepatitis C virus (HCV) reference panel provided a coding-genome example spanning seven recognized genotypes [16]. Every complete gene and codon-position partition retained signal, but 36-site Z-score curves revealed localized losses (Fig. 6). Across genotype 1, 2, and 3 branches, original SatuTe classified 1/507, 4/507, and 5/507 position-1+2 windows as saturated, compared with 11/252, 34/252, and 33/252 third-position windows. Eigenvalue weighting reduced the corresponding third-position counts to 6, 24, and 27 (Supplementary Table S6).

A gene-level screen would therefore retain every locus, whereas SatuTe profiles reveal signal loss associated with third positions and specific intervals. These profiles can motivate partition-specific model checks or selective masking without requiring wholesale removal of otherwise informative genes.

Discussion

SatuTe asks whether an alignment retains detectable dependence between the subtrees separated by a branch. This question differs from bootstrap and likelihood-ratio support, which quantify the reproducibility of, or preference for, a split [1, 2, 17]. The original JC benchmark makes the distinction explicit: bootstrap support can remain complete after SatuTe reaches its null rejection rate. A non-significant SatuTe result therefore indicates insufficient branch-specific signal in the tested alignment and model; it does not reject common ancestry.

The original SatuTe formulation remains the primary model in this study. It supplies the null hypothesis, branch-specific interpretation, one-sided decision rule, and baseline simulation design. The eigenvalue-weighted statistic is a model-aware extension that preserves each of these elements. It reduces exactly to original SatuTe under JC but allows faster non-stationary directions to contribute while they persist under GTR or amino-acid models.

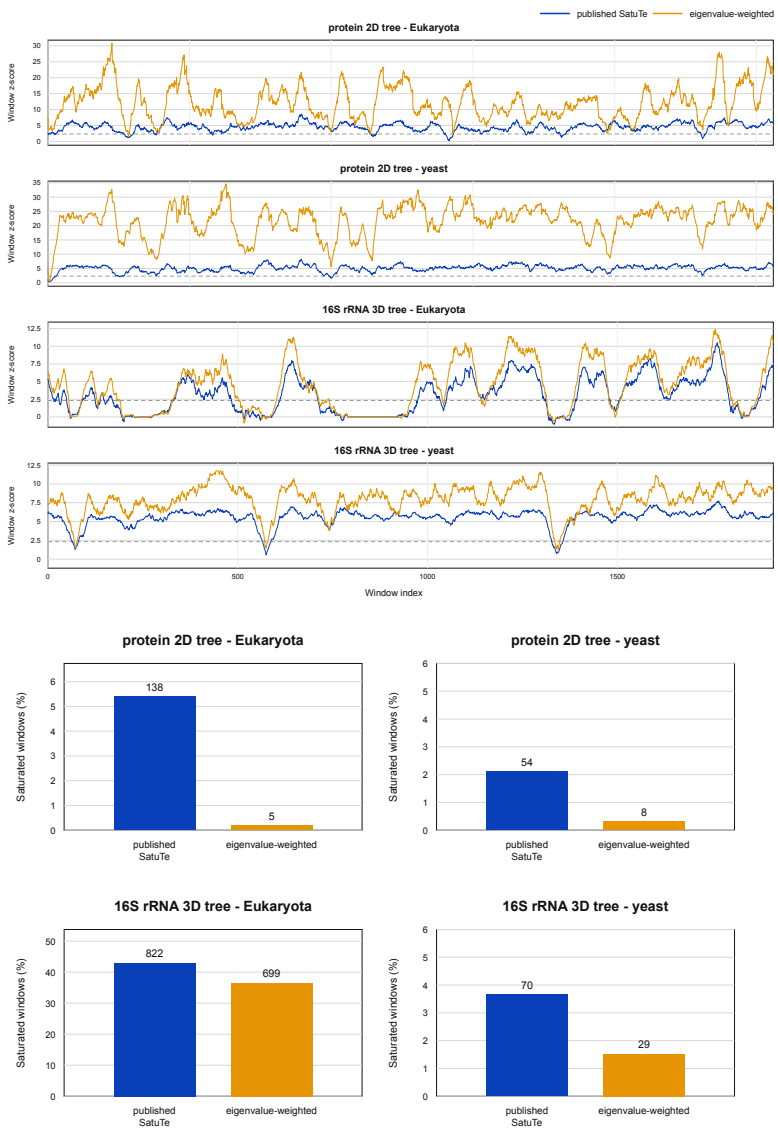


Figure 5: **Branch-specific signal across Tree-of-Life alignments. a,** Thirty-six-site window Z-scores for Eukaryota and yeast branches in protein 2D and rRNA 3D trees; the dashed line marks the decision threshold. **b,** Fractions and counts of saturated windows. Weighting changes threshold crossings most strongly for the rRNA Eukaryota branch but preserves the protein–rRNA contrast established by original SatuTe.

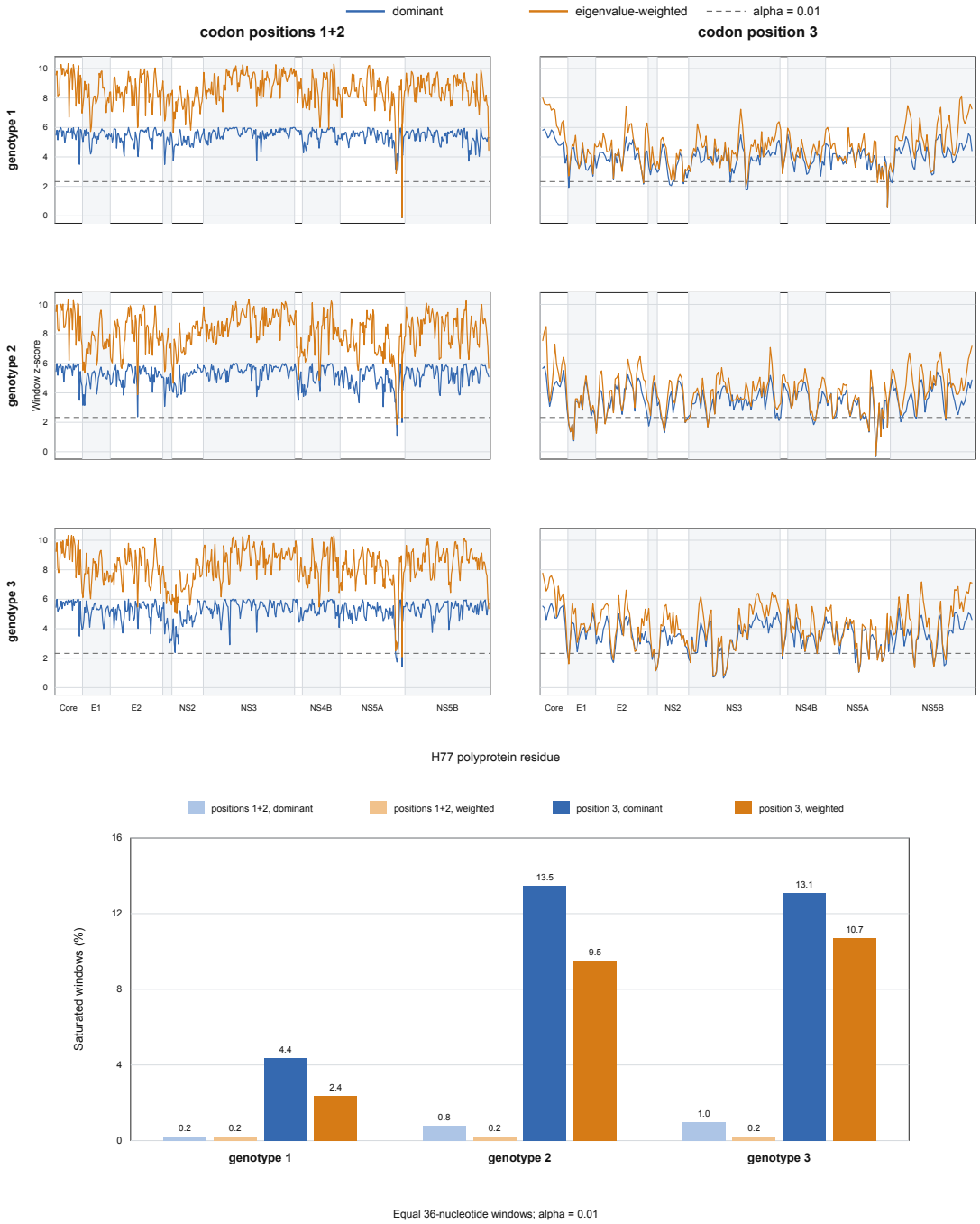


Figure 6: **Codon-position localization in an HCV reference panel. a,** Original and eigenvalue-weighted SatuTe Z-scores along the H77 polyprotein coordinate for fixed-topology 36-site windows. The dashed line marks the descriptive $\alpha = 0.01$ threshold. **b,** Fractions of saturated windows. Losses concentrate in the synonymous-enriched third-position partition; weighting reduces, but does not remove, the pattern concealed by whole-gene tests.

The benefit of weighting was localized rather than universal. Gains occurred mainly in short alignments near the long-branch detection boundary and were negligible when both formulations had high power or had reached the null floor. Branch-neighborhood audits showed that weighting did not indiscriminately increase scores across a tree. The size of the effect also depended strongly on the substitution matrix: JTT produced broader shifts than LG or Q.pfam, showing that alphabet size alone does not determine the available spectral information.

Model adequacy was at least as consequential as the choice of statistic. JC and K2P analyses of GTR data were too liberal, whereas F81 was closer to the correctly specified result. The JC zero-difference control establishes equality between the original and weighted formulas under JC; it does not establish the adequacy of JC for data generated under a different process.

Invariant sites create a separate structural problem. A zero-rate category remains ancestrally dependent across an infinite focal branch. Placing that category in a product-form null therefore introduces a positive mean that studentization amplifies as alignment length increases. Including the invariant category's joint probability only in the null denominator restored exact centering and standard-normal calibration. After independent empirical thresholding, gains again followed the detection boundary. The exploratory $+I+R4$ analysis serves as a stress test, whereas fixed GTR+F+I+G4 is the publication benchmark.

The empirical examples demonstrate the value of a local diagnostic. Ribosomal proteins retained signal across nearly all windows, whereas 16S rRNA contained broad saturated regions and lost signal for the Eukaryota branch in its fastest category. In HCV, whole-gene tests were uniformly informative, yet short-window losses concentrated at synonymous-enriched third positions. An alignment-wide pass can therefore conceal within-locus saturation, whereas SatuTe profiles can guide partition-specific model checks or selective masking without discarding an entire coding region. Weighting reduced the number of threshold crossings without changing either biological contrast.

These examples also define the limits of interpretation. SatuTe localizes signal but does not select the correct topology, establish model adequacy, or estimate viral population diversity. The HCV analysis is further limited by its ten-genome reference panel, overlapping short windows, and model refitting at that scale. Non-stationary or non-reversible extensions would make the root identifiable and require a root-aware null.

Within the present reversible framework, we recommend reporting original SatuTe as the primary branch-specific diagnosis, accompanied where appropriate by weighted sensitivity results, paired differences, multiplicity correction, conventional branch support, and model-adequacy checks. Integrated into IQ-TREE, this reporting structure makes the loss of historical signal testable at the level where phylogenetic conclusions are drawn: individual branches and alignment regions.

Online Methods

Notation

A focal branch AB separates subtrees T_A and T_B. A site pattern ∂ induces subpatterns ∂_A and ∂_B, with partial likelihood vectors $L(\partial_A)$ and $L(\partial_B)$ at the branch endpoints. The stationary distribution is π, and the subtree-pattern probabilities are $P(\partial_A) = \langle L(\partial_A), \pi \rangle$ and $P(\partial_B) = \langle L(\partial_B), \pi \rangle$. The reversible rate matrix has eigenvalues $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$ and left eigenvectors h_i, where d is the number of non-stationary components. Coherence coefficients are defined in Eq. (1).

Original SatuTe estimator

Original SatuTe uses the coherence coefficient associated with the slowest non-stationary component, including the full eigenspace when that eigenvalue is repeated. For n independently

evolving sites, it averages the sitewise coefficients and divides the resulting mean by its estimated standard error. The one-sided statistic rejects subtree independence when $Z > z_\alpha$. For a topology inferred from the tested alignment, the taxon-pair Bonferroni threshold is $z_{\alpha/(m_A m_B)}$, where m_A and m_B are the numbers of taxa in the two subtrees.

Eigenvalue-weighted estimator

For each site s, $C_{\partial s i}$ is computed for every non-stationary mode. At branch length t, mode i receives weight $w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}$. The quantities $\hat{C}_\lambda(t)$, $\hat{\sigma}^2_\lambda(t)$, and $Z_\lambda(t)$ are then calculated using Eqs. (4) to (6). If the slowest non-stationary eigenvalue is repeated, every direction in that eigenspace has weight one. Under JC, the estimator is identical to original SatuTe.

Null distribution

The weighted statistic retains the SatuTe null hypothesis. Let

$$X_i(\partial_A) = \left\langle \frac{L(\partial_A)}{P(\partial_A)}, h_i \right\rangle, \qquad Y_i(\partial_B) = \left\langle \frac{L(\partial_B)}{P(\partial_B)}, h_i \right\rangle.$$

Then $C_{\partial i} = X_i Y_i$ and

$$C^\lambda_\partial(t) = \sum_{i=1}^{d} w_i(t) X_i Y_i.$$

Under the SatuTe null hypothesis, ∂_A and ∂_B are independent. Because the non-stationary eigenvectors are orthogonal to the stationary component, the original SatuTe argument gives $\mathbb{E}[X_i] = 0$ and $\mathbb{E}[Y_i] = 0$. Therefore $\mathbb{E}[C_{\partial i}] = 0$ for every non-stationary component and, by linearity, $\mathbb{E}[C^\lambda_\partial(t)] = 0$.

The variance follows by expanding the square of the weighted sum. Define the subtree second moments

$$\sigma^2_{A,i,j} = \mathbb{E}[X_i X_j], \qquad \sigma^2_{B,i,j} = \mathbb{E}[Y_i Y_j].$$

Independence of the two subtrees under the null gives

$$\mathrm{Var}\Big(C^\lambda_\partial(t)\Big) = \sum_{i=1}^{d}\sum_{j=1}^{d} w_i(t) w_j(t) \sigma^2_{A,i,j} \sigma^2_{B,i,j}. \tag{12}$$

For n independent sites, $\mathrm{Var}[\hat{C}_\lambda(t)] = \mathrm{Var}(C^\lambda_\partial(t))/n$, which is the population form estimated in Eq. (5). With finite fourth moments and a consistent variance estimator, the central limit theorem and Slutsky's theorem give $Z_\lambda(t) \xrightarrow{d} N(0,1)$ under subtree independence. Maximum-likelihood topology selection remains a separate model-selection step, so we retain the Bonferroni correction used by original SatuTe for that scenario.

Simulation protocol

The five-taxon tree is a balanced four-taxon subtree T_A, with all subtree branches set to 0.2, and a single taxon B attached by focal branch AB. The 16-taxon tree contains two balanced eight-taxon subtrees; internal and external lengths are drawn from EvoNAPS-derived distributions and restricted to 0.04–0.2. The original SatuTe baseline uses JC, $n \in \{100, 1{,}000, 10{,}000\}$, 1,000 replicates per cell, and focal lengths

$$t \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.8, 1, 1.5, 2, 2.5, 3, 3.5, 4, 5, 7.5, 10\}.$$

Every alignment is analyzed under four scenarios: the true topology and branch lengths; the true topology with ML branch lengths; an ML-inferred topology and branch lengths; and

that inferred-tree analysis with threshold $\alpha/(m_A m_B)$. For the five- and 16-taxon targets, the Bonferroni denominators are 4 and 64, respectively. Baseline alignments were simulated with Seq-Gen and evaluated with IQ-TREE and SatuTe [18, 4].

The rate-heterogeneous extensions use full alignments and paired formula evaluation. Within each replicate, original and weighted formulas use the same alignment, tree, branch length, target split, and random seed. This pairing isolates the formula effect from simulator and topology variation.

The nucleotide simulations use the Tree-of-Life 16S rRNA GTR+F+G4 setting, implemented as a fixed four-category model in the IQ-TREE integration. Exploratory protein simulations use the Tree-of-Life LG+G4 setting. The growing-site grid uses

$$n \in \{100, 250, 500, 1{,}000, 2{,}000, 4{,}000, 10{,}000\},$$

focal lengths $4, 5, 6, 7.5, 10, 12$, and 100 replicates per cell. The corrected publication suite uses 1,000 replicates, alignment lengths of $100, 250, 1{,}000$ sites, and focal lengths $4, 5, 6, 7, 8, 9, 10, 12$. It evaluates fixed GTR+F, GTR+F+G4, GTR+F+I+G4, LG+G4, WAG+G4, JTT+G4, and Q.pfam+G4 models. The branch-neighborhood audit computes both statistics for the target and adjacent branches in the same trees.

For model misspecification, alignments are generated under fixed GTR+F and evaluated under GTR, JC, K2P, or F81. This comparison separates formula equality under JC from the adequacy of JC for GTR-generated data. We report formula-specific informative-call fractions in addition to paired differences.

The independent calibration experiment uses 5,000 exact-null replicates in every cell, with replicates 1–2,500 prespecified for threshold training and replicates 2,501–5,000 for held-out evaluation. A separate 2,000-replicate design alternates target-null and target-alternative branches to form a 50:50 mixed-null sample for empirical false-discovery rate and power. For discrete adjusted p-values, the training-derived cutoff is the largest attainable value for which the complete tied block has empirical probability at most 0.05; if the minimum value already exceeds that mass, the cutoff is placed immediately below it. This tie-aware definition makes the fitted cutoff agree with the implemented $p \leq c$ decision rule. We report Wilson intervals for held-out null rejection and paired bootstrap intervals for differences between formulas.

All publication outputs use schema-v2 detail, summary, branch-audit, decision-rule-audit, and timing tables. Replicate completeness is checked within every model–tree–scenario–site–branch cell before aggregation. Paired percentile-bootstrap intervals resample replicate-level discordance, and exact McNemar tests use the two discordant call counts.

Paired invariant-mixture grid

The $+I$ analysis uses a four-taxon tree with two two-tip subtrees, terminal branches of length 0.05 and a varied focal branch. The substitution process is fixed to GTR with exchangeabilities $\{1, 2, 1, 1, 2, 1\}$ and stationary frequencies $\{0.30, 0.20, 0.20, 0.30\}$. Both models have invariant proportion $q_0 = 0.10$. The $+I + G4$ model uses four equal-probability discrete-gamma categories with shape 0.5. The $+I + R4$ model uses positive-category proportions $\{0.35, 0.25, 0.23, 0.17\}$ and rates $\{0.0747656, 0.420849, 1.15850, 3.54214\}$, rescaled after addition of the invariant component so that the overall mean rate is one.

The $+I + R4$ construction is retained only as an exploratory control. The publication benchmark instead uses the fixed $+I + G4$ specification. Fixed-model GTR+F+G4 and GTR+F+I+G4 comparisons are reported separately.

All $4^4 = 256$ site-pattern probabilities are enumerated exactly under the correct finite-branch distribution and under the infinite-branch null in Eq. (9). Each simulated alignment is a multinomial draw from one of these exact distributions. The same pattern counts are evaluated by the counterfactual and refined statistics, producing paired decisions without simulator-stream variation.

Alignment lengths are

$$n \in \{100, 500, 1000, 5000, 10000\}. \tag{13}$$

The dense focal-branch grid is

$$t \in \{0.5, 1, 2, 3, 4, 5, 7.5, 10, 12.5, 15, 17.5, 20, 22.5, 25, \\ 30, 35, 40, 45, 50, 60, 75, 100, 150, 200, 500, 1000, 10000\}. \tag{14}$$

Thus the grid uses 2.5-unit resolution from 10 to 25 and 5-unit resolution from 30 to 50, where the detection transitions occur, while retaining short and extreme controls. For every model, branch length and alignment length, 20,000 null and 20,000 finite-branch alternative alignments are drawn. Half of the null replicates train a formula-specific empirical 95th-percentile threshold; the other half evaluates held-out false-positive control. Power is measured on the independent alternative sample. Wilson intervals quantify individual rejection fractions. Paired percentile-bootstrap intervals use 4,000 resamples of the three discordance outcomes: refined only, agreement and counterfactual only.

Biological sliding-window analyses

The protein 2D and rRNA 3D examples use the alignments and topologies of Hug and colleagues [12]. Branch lengths are re-estimated in IQ-TREE under LG+G4 for the 2,596-site, 3,083-taxon protein alignment and GTR+G4 for the 1,947-site, 1,871-taxon rRNA alignment. Each site is assigned to the rate category with largest posterior probability. Whole-category branch tests use the taxon-pair Bonferroni threshold because a category contains roughly one quarter of the alignment.

Sliding windows contain 36 sites, the minimum scale used in the original analysis. A global variance estimate is formed from the category-specific estimates weighted by their full-alignment site fractions; each window uses its mean coherence and this global variance. The original analysis does not apply Bonferroni correction to windows because one window contains less than 2% of the alignment used to infer the tree. Saturated windows satisfy $Z \leq z_\alpha$. Hypervariable rRNA regions V1–V9 and conserved regions C1–C9 are mapped from bacterial 16S annotations after accounting for alignment gaps [15, 19].

The curated rRNA rerun compares published SatuTe, a native dominant recomputation and a native weighted recomputation for the Eukaryota and yeast branches. The independent rerun reconstructs site scores from the IQ-TREE treefile, model report, site-probability output and original alignments for both Tree-of-Life datasets. Cross-implementation count differences are reported descriptively and are not treated as paired formula effects.

For the topology-rearrangement analysis, the Eukaryota subtree is pruned from the protein 2D tree and regrafted onto the Bacteria–Archaea branch to create a 3D alternative; the reverse operation creates a 2D alternative from the rRNA 3D tree. Branch lengths are re-estimated under the corresponding LG+G4 or GTR+G4 model. Protein-specific scores are calculated for each of 16 concatenated proteins, and rRNA-specific scores for nine annotated regions. Differences are defined consistently between 2D and 3D placements and compared descriptively, with a two-sample Kolmogorov–Smirnov test for the two empirical difference distributions.

For the HCV example, complete accession-versioned references were selected for subtypes 1a, 1b, 2a, 2b, 3a, 3b, 4a, 5a, 6a and 7a. Polyprotein translations were aligned with MAFFT 7.526 and back-translated; H77 mature-peptide annotations defined Core–NS5B. A full-coding topology was inferred under GTR+F+G4, and the subtype pairs for genotypes 1–3 defined target branches. Codon positions 1+2 and position 3 were concatenated separately. Fixed-topology 36-site windows advanced by 12 sites, with model parameters and branch lengths re-estimated per window. Positions 1+2 and position 3 therefore had equal statistical sample size but spanned 18 and 36 aligned codons.

For every HCV window we recorded both the dominant and eigenvalue-weighted standardized statistics and plotted them at the corresponding H77 polyprotein midpoint. The horizontal decision line is $z_{0.01} = 2.326$; a window is counted as saturated when its Z-score falls below this line. The 507 positions-1+2 windows and 252 position-3 windows were required for each of three target branches and both formulas, yielding a prespecified 4,554-row grid. We did not correct across overlapping windows because these curves are descriptive localization profiles. The ten-reference design tests partition localization, not population diversity, recombination or genotype assignment.

Data availability

Source data, processed aggregate tables and figure-source files will be deposited in a versioned public archive before publication. The Tree-of-Life alignments are available from the studies in which they were published [12, 13]. HCV records are available from NCBI Nucleotide under accession versions NC_004102.1, AJ238799.1, AB047639.1, D10988.1, D17763.1, D49374.1, Y11604.1, AF064490.1, Y12083.1 and EF108306.2; the accession bundle, alignments, complete 4,554-row window grid and figures are retained in the versioned HCV release. Raw simulation shards are retained on the Vienna Life Science Compute Cluster and are available from the corresponding author on reasonable request. The archival DOI and a reviewer-access link will be added before submission.

Code availability

The development repository is hosted at `https://github.com/EnesSakalliUniWien/iqtree3`. The IQ-TREE integration, simulation and postprocessing scripts, plotting code and manuscript sources used for this study will be archived with the source data before submission. Exact source versions are recorded in the experiment and run manifests.

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B.S. conceived the extension, developed the statistical formulation, performed the analyses and wrote the manuscript.

Competing interests

The author declares no competing interests.

Additional information

Further information on research design is available in the Nature Portfolio Reporting Summary associated with this article.